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Safety System for Valuable Objects

Abstract

The purpose of this project is to integrate our basic skills on VHDL and logic design concepts. In this experiment, I will recreate a system that is being used in museums or exhibitions. Since the safety of such valueable objects is a huge concern, I believe this topic worths to work on it. This project will be BASYS3 based written in VHDL and will cover EEE102 course material.

Design Specification

First of all, BASYS 3 board will be used. To implement the system, firstly, PIR (passive infrared) motion sensor will be used to detect outer people and actions around the object. It will be connected to Pmod (Peripheral Module) port of the board. These ports can be used as simple expansion ports, since all of the pin-outs correspond to pins on the FPGA.[1] Also, if needed, to secondary voltage and ease connections, Arduino Uno and a breadboard to connect this will be used. Finally, after the detection, in order for the system to work, a buzzer will be implemented and it will sound according to the current state. And I will also control the room with a smoke sensor.

[1] Digilent, "Basys 3 Reference Manual," accessed March 08, 2023. https://digilent.com/reference/programmable-logic/basys-3/reference-manual#fpga_configurations

Equipments

- BASYS 3 board
- PIR motion sensor
- Arduino Uno
- Breadboard, jumper cables
- Buzzer
- Smoke Sensor

Methodology

This system will provide safety in surroundings of an object. PIR sensors have two slots, one of them is sensitive to infrared lights.Their working system basically depends on the change of heat that all objects emit. When there is a change in heat, for instance an animal or human passes by, there is a positive or negative voltage change between two slots. With this voltage difference, it allows to detect motion in its vision. In order to make safer connections between sensors and FPGA in real life setup and to supply voltage (if needed another voltage supply than BASYS 3 supplies), I am planning to use an Arduino Uno. This is a board which is commonly used by electronics engineers or people interested in electronics. It supplies 5V and highly practical due to its ports and portability. Lastly, the signals created by PIR sensor will be used by buzzer and it will sound while there is a motion in vision. Buzzers are components that are aimed to make sound. Magnetic buzzers work by creating magnetic fields and vibrating a ferromagnetic disk, while, piezo buzzers are based on piezo electric concept.If a signal comes to buzzer, the ceramic inside it vibrates rapidly and this creates sound. [2] And lastly I will connect a smoke sensor for possible fires or smoke that can damage materials aimed to be protected.

After this technical part, I will implement a real life mechanism for this safety system. I will form a box, symbolizing a room of a museum or exhibition. On the corner of this box, there will be an object protected and exhibited. The sensor will be located on the top of the object and rest of the system will be hidden above the ceiling or behind the box.

In Demo 1, I will connect PIR sensor to BASYS 3 board and I will code and test the system according to the project plan. I also will connect buzzer. Furthermore, on Final Demo, I will connect smoke sensor to the system and I will prepare visual, non-technical setup such as box and environment symbolizing a museum and the object symbolizing valuable object.

[2] American Piezo, “Piezo Buzzers vs. Magnetic Buzzers — What’s the Difference?,” accessed on March 08, 2023. https://www.americanpiezo.com/blog/piezo-buzzers-vs-magnetic-buzzers/#:~:text=The%20flexible%20ferromagnetic%20disk%20is,movement%20makes%20the%20buzzer%20sound

References

Digilent. “Basys 3 Refenrence Manual.” Accessed on March 08, 2023. https://digilent.com/reference/programmable-logic/basys-3/reference-manual#fpga_configurations

American Piezo. “Piezo Buzzers vs. Magnetic Buzzers — What’s the Difference?” Accessed on March 08, 2023. https://www.americanpiezo.com/blog/piezo-buzzers-vs-magnetic-buzzers/#:~:text=The%20flexible%20ferromagnetic%20disk%20is,movement%20makes%20the%20buzzer%20sound